

Boss Challenge — Paper 06 (Class 7)

Soil | Forests: Our Lifeline | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The dark, top layer of soil that is richest in humus and best for plant growth is the:

- (A) bedrock
- (B) subsoil
- (C) topsoil
- (D) the sky

2. In a forest, green trees and plants that make their own food are called:

- (A) decomposers
- (B) predators
- (C) consumers
- (D) producers

3. The ratio 4 : 6 in its simplest form is:

- (A) 4 : 6
- (B) 2 : 3
- (C) 3 : 2
- (D) 1 : 2

4. Soil is formed slowly by the breaking down of:

- (A) water only
- (B) plastic
- (C) rocks (weathering)
- (D) air alone

5. Animals that eat only plants are called:

- (A) producers
- (B) herbivores
- (C) carnivores
- (D) decomposers

6. 25% of 200 is:

- (A) 75
- (B) 25
- (C) 50
- (D) 100

7. Soil through which water passes very quickly (high percolation) is mainly:

- (A) clayey soil
- (B) sandy soil
- (C) rock
- (D) loamy soil

8. Tiny living things (like fungi and bacteria) that break down dead leaves and animals are:

- (A) producers
- (B) decomposers
- (C) carnivores
- (D) herbivores

9. $\frac{1}{2}$ written as a percentage is:

- (A) 5%
- (B) 100%
- (C) 2%
- (D) 50%

10. Clayey soil is the best choice for:

- (A) growing most crops
- (B) fast drainage
- (C) a beach surface
- (D) making pots and bricks

11. Why are forests often called the 'green lungs' of the Earth?

- (A) they make plastic
- (B) they give out CO $\blacksquare$
- (C) they store water only
- (D) they take in carbon dioxide and give out oxygen

12. 0.75 written as a percentage is:

- (A) 57%
- (B) 0.75%
- (C) 7.5%
- (D) 75%

13. The decayed remains of plants and animals that make soil fertile are called:

- (A) clay
- (B) sand
- (C) bedrock
- (D) humus

14. A simple food chain in a forest could be:

- (A) grass -> tiger -> deer
- (B) grass -> deer -> tiger
- (C) deer -> grass -> tiger
- (D) tiger -> deer -> grass

15. 20% written as a fraction in lowest terms is:

- (A) $\frac{1}{20}$
- (B) $\frac{1}{5}$
- (C) $\frac{2}{10}$
- (D) $\frac{5}{1}$

16. Loamy soil is best for growing most crops because it:
- (A) holds no water
 - (B) holds enough water AND lets air and roots through
 - (C) never lets water drain
 - (D) has no nutrients
17. Which best describes why forests help PREVENT floods and soil erosion?
- (A) tree roots hold the soil and slow down rainwater
 - (B) leaves block all rain
 - (C) trees push water uphill
 - (D) forests make it rain less
18. 15 out of 25 students passed. The pass percentage is:
- (A) 15%
 - (B) 40%
 - (C) 60%
 - (D) 75%
19. Which property is HIGHEST in clayey soil?
- (A) particle size
 - (B) air content
 - (C) percolation rate
 - (D) water-holding capacity
20. Removing large areas of forest (deforestation) can lead to:
- (A) cleaner air
 - (B) more soil erosion and less rainfall
 - (C) richer soil
 - (D) more oxygen
21. A shopkeeper buys a toy for ■100 and sells it for ■120. His profit percent is:
- (A) 20
 - (B) 12%
 - (C) 20%
 - (D) 120%
22. The hardest, deepest layer beneath the soil is the:
- (A) subsoil
 - (B) topsoil
 - (C) bedrock
 - (D) humus
23. In a forest, the topmost layer formed by the branches of the tallest trees is the:
- (A) understorey
 - (B) bedrock
 - (C) forest floor
 - (D) canopy

24. A bag costs ■200 and is sold for ■150. The loss percent is:
- (A) 50%
 - (B) 25
 - (C) 25%
 - (D) 33%
25. Why does sandy soil feel dry soon after rain?
- (A) water percolates (drains) through it quickly
 - (B) it absorbs no water at all
 - (C) it holds water tightly
 - (D) it makes water
26. Decomposers are important in a forest because they:
- (A) eat living animals
 - (B) hunt deer
 - (C) make oxygen
 - (D) return nutrients from dead matter back to the soil
27. Simple interest on ■1000 at 5% per year for 2 years is:
- (A) ■200
 - (B) ■1100
 - (C) ■50
 - (D) ■100
28. Which sequence is the correct soil profile from top to bottom?
- (A) topsoil -> subsoil -> weathered rock -> bedrock
 - (B) bedrock -> topsoil -> subsoil
 - (C) sand -> clay -> water
 - (D) subsoil -> topsoil -> bedrock
29. Which is TRUE about a forest?
- (A) only trees live there
 - (B) it has no soil
 - (C) nothing depends on anything
 - (D) it is a system where plants, animals and decomposers depend on each other
30. If 5 pens cost ■50, then 1 pen costs:
- (A) ■250
 - (B) ■50
 - (C) ■10
 - (D) ■5
31. A farmer waters a field and most water drains away within minutes. The soil is most likely:
- (A) loamy
 - (B) sandy
 - (C) rocky and solid
 - (D) clayey

- 32.** Forests help maintain the water cycle mainly by:
- (A) drinking up all the rivers
 - (B) making the air dry
 - (C) stopping evaporation
 - (D) releasing water vapour (transpiration) and helping rainfall
- 33.** 30% of 50 is:
- (A) 20
 - (B) 15
 - (C) 30
 - (D) 35
- 34.** Adding rotted leaves and cow dung (compost) to soil mainly increases its:
- (A) dryness
 - (B) humus and fertility
 - (C) rockiness
 - (D) sand content
- 35.** Animals that hunt and eat other animals are called:
- (A) decomposers
 - (B) herbivores
 - (C) carnivores
 - (D) producers
- 36.** A price of ₹200 is increased by 25%. The new price is:
- (A) ₹175
 - (B) ₹250
 - (C) ₹50
 - (D) ₹225
- 37.** Which soil has the SMALLEST particles and the slowest percolation?
- (A) gravel
 - (B) sandy soil
 - (C) clayey soil
 - (D) loamy soil
- 38.** Dead leaves on the forest floor are important because they:
- (A) decay into humus that enriches the soil
 - (B) poison the soil
 - (C) are useless waste
 - (D) block all sunlight forever
- 39.** Express the ratio 50 cm to 1 m in simplest form.
- (A) 1 : 1
 - (B) 1 : 2
 - (C) 50 : 1
 - (D) 2 : 1

40. Topsoil is dark in colour mainly because it contains:
- (A) humus (decayed organic matter)
 - (B) plastic
 - (C) only sand
 - (D) water only
41. Which group is at the START of every forest food chain?
- (A) green plants (producers)
 - (B) the Sun
 - (C) decomposers
 - (D) tigers
42. $\frac{3}{5}$ written as a percentage is:
- (A) 53%
 - (B) 60%
 - (C) 35%
 - (D) 3%
43. Which is the best soil for growing paddy (rice), which needs standing water?
- (A) sandy soil
 - (B) clayey soil (holds water)
 - (C) gravel
 - (D) dry rock
44. Why is biodiversity (many kinds of plants and animals) important in a forest?
- (A) it keeps the forest balanced and able to recover from harm
 - (B) it makes the forest weaker
 - (C) it stops all growth
 - (D) it is useless
45. A marked price of ₹500 has a 10% discount. The amount to pay is:
- (A) ₹450
 - (B) ₹550
 - (C) ₹50
 - (D) ₹490
46. Soil erosion (the washing or blowing away of topsoil) is BEST reduced by:
- (A) planting trees and grass to hold the soil
 - (B) removing all plants
 - (C) paving everything
 - (D) watering more
47. A food WEB is different from a food chain because it:
- (A) has only one path
 - (B) ignores animals
 - (C) shows many linked food chains together
 - (D) has no plants

48. 12.5% as a fraction in lowest terms is:

- (A) $\frac{1}{8}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{12}$
- (D) $\frac{1}{4}$

49. Which list shows soils in order of INCREASING particle size?

- (A) clay, silt, sand
- (B) sand, clay, silt
- (C) sand, silt, clay
- (D) silt, sand, clay

50. Which is a reason forests are called our 'lifeline'?

- (A) they have no use
- (B) they only give shade
- (C) they give oxygen, food, wood, medicines and shelter to many living things
- (D) they only block roads

51. If $a : b = 2 : 3$ and $a = 8$, then b is:

- (A) 6
- (B) 3
- (C) 12
- (D) 16

52. Which is NOT a use of soil?

- (A) growing crops
- (B) making bricks and pots
- (C) generating electricity in a power station
- (D) a home for earthworms

53. If all the tigers (top carnivores) in a forest disappeared, a likely result is:

- (A) nothing would change
- (B) decomposers would vanish
- (C) the number of deer would rise sharply, over-grazing the plants
- (D) plants would increase

54. Increase 80 by 25%. The result is:

- (A) 105
- (B) 60
- (C) 20
- (D) 100

55. Which soil would you choose to make a strong clay pot?

- (A) clayey soil
- (B) gravel
- (C) loamy soil
- (D) sandy soil

56. Which of these is the correct flow of energy in a forest?
- (A) plants → Sun → animals
 - (B) carnivores → herbivores → plants
 - (C) Sun → plants → herbivores → carnivores
 - (D) herbivores → plants → Sun
57. What percent is 40 of 160?
- (A) 400%
 - (B) 12.5%
 - (C) 40%
 - (D) 25%
58. Air present in the spaces between soil particles is important because it:
- (A) makes the soil red
 - (B) is needed by roots and soil organisms to breathe
 - (C) is useless
 - (D) stops plants growing
59. Humus in forest soil is produced by:
- (A) decomposers breaking down dead leaves and animals
 - (B) rain washing rocks
 - (C) sunlight alone
 - (D) tigers hunting deer
60. $A : B = 3 : 4$ and $B : C = 4 : 5$. Then $A : C$ is:
- (A) 4 : 5
 - (B) 3 : 4
 - (C) 5 : 3
 - (D) 3 : 5
61. Why do plants struggle to grow in pure sandy soil?
- (A) it has too much humus
 - (B) water and nutrients drain away too fast
 - (C) it is too sticky
 - (D) it holds too much water
62. Forests are important for wildlife because they provide:
- (A) only roads
 - (B) nothing useful
 - (C) only firewood
 - (D) food and shelter (habitat) for many animals
63. A student scored 18 out of 20 in a test. The percentage is:
- (A) 9%
 - (B) 80%
 - (C) 90%
 - (D) 18%

64. Which step helps STOP soil from being washed away on a sloping hill?

- (A) removing all plants
- (B) digging the slope bare
- (C) pouring more water
- (D) making terraces (steps) and planting trees

65. The 'understorey' of a forest is the layer:

- (A) of dead leaves on the ground
- (B) of shrubs and small plants below the tall-tree canopy
- (C) of the tallest trees
- (D) deep underground

66. The simple interest on ■2000 at 10% per year for 3 years is:

- (A) ■600
- (B) ■200
- (C) ■2600
- (D) ■60

67. Percolation rate of water is HIGHEST in:

- (A) loamy soil
- (B) sandy soil
- (C) clayey soil
- (D) solid rock

68. Why should we protect and not over-cut forests?

- (A) they support life by giving oxygen, rain, soil and habitats
- (B) they waste land
- (C) they are dangerous
- (D) they block sunlight

69. If 60% of a number is 30, the number is:

- (A) 50
- (B) 18
- (C) 48
- (D) 90

70. Which best describes weathering in soil formation?

- (A) rocks suddenly exploding
- (B) rivers freezing
- (C) planting seeds
- (D) the slow breaking of rocks by sun, water, wind and living things

71. A forest acts like a giant sponge during heavy rain because:

- (A) it makes the rain stop
- (B) trees drink all the water
- (C) its soil and leaf litter soak up and slowly release water
- (D) it sends water to space

72. A number is increased from 50 to 60. The percentage increase is:
- (A) 16.7%
 - (B) 20%
 - (C) 120%
 - (D) 10%
73. The layer just below the topsoil, lighter in colour with less humus, is the:
- (A) humus
 - (B) subsoil
 - (C) topsoil
 - (D) bedrock
74. Which pair correctly matches the organism with its role?
- (A) tiger - carnivore (consumer)
 - (B) mushroom - producer
 - (C) grass - decomposer
 - (D) deer - carnivore
75. Two ratios 2 : 3 and 4 : 6 are:
- (A) opposite
 - (B) not equal
 - (C) impossible to compare
 - (D) equal (in proportion)
76. Which best explains why earthworms are called a 'farmer's friend'?
- (A) they make it rain
 - (B) they eat crops
 - (C) they loosen and mix the soil, letting in air and water
 - (D) they poison pests
77. Which human action most directly HARMS forests and their food webs?
- (A) watering plants
 - (B) studying animals
 - (C) cutting down large numbers of trees (deforestation)
 - (D) planting more trees
78. 40% of a class of 50 are girls. The number of girls is:
- (A) 20
 - (B) 40
 - (C) 10
 - (D) 25
79. Loamy soil is a mixture of:
- (A) only clay
 - (B) only rock pieces
 - (C) sand, clay and humus in good proportion
 - (D) only sand

80. Forests are sometimes called 'carbon sinks' because they:
- (A) have no carbon
 - (B) give out carbon dioxide
 - (C) burn carbon for fuel
 - (D) take in carbon dioxide and store carbon in wood and soil
81. A man earns ■12000 a month and saves 15% of it. His monthly saving is:
- (A) ■1800
 - (B) ■1500
 - (C) ■180
 - (D) ■15000
82. If you want the soil in a pot to drain quickly so roots don't rot, you would add some:
- (A) plastic
 - (B) sand
 - (C) more clay
 - (D) oil
83. Which statement about forests is FALSE?
- (A) forests give out carbon dioxide and take in oxygen overall
 - (B) forests support the water cycle
 - (C) forests are habitats for wildlife
 - (D) forests help prevent soil erosion
84. The price of a book rises from ■80 to ■100. The percentage increase is:
- (A) 25
 - (B) 20%
 - (C) 25%
 - (D) 80%
85. Which is the MAIN reason topsoil is so valuable to farmers?
- (A) it is the hardest layer
 - (B) it has no life
 - (C) it drains all water away
 - (D) it is rich in humus and nutrients for crops
86. Cutting a forest can reduce local rainfall because fewer trees means:
- (A) rivers flow faster forever
 - (B) more clouds form
 - (C) less water vapour released into the air
 - (D) the Sun gets hotter
87. A toy marked ■250 is sold at a 20% discount. The selling price is:
- (A) ■230
 - (B) ■300
 - (C) ■200
 - (D) ■50

88. Which natural agents are MOST responsible for soil erosion?
- (A) humus
 - (B) running water and wind
 - (C) earthworms
 - (D) sunlight only
89. Which best shows that the parts of a forest are INTERDEPENDENT?
- (A) plants feed animals, animals spread seeds, and decomposers enrich soil for plants
 - (B) every plant lives alone
 - (C) animals never help plants
 - (D) soil needs nothing
90. If 3 kg of apples cost ₱150, the cost of 5 kg (same rate) is:
- (A) ₱250
 - (B) ₱50
 - (C) ₱90
 - (D) ₱450
91. Why is studying the soil profile (its layers) useful?
- (A) it has no use
 - (B) it tells farmers and builders about the soil's depth, fertility and how it holds water
 - (C) it shows the weather
 - (D) it predicts earthquakes
92. The main job of the forest canopy (the leafy roof of tall trees) is to:
- (A) catch most of the sunlight for photosynthesis
 - (B) hold the soil together
 - (C) store water underground
 - (D) make bricks
93. What percent is 60 of 240?
- (A) 60%
 - (B) 25%
 - (C) 4%
 - (D) 40%
94. Which soil feels gritty and rough when rubbed between the fingers?
- (A) sandy soil
 - (B) loamy soil
 - (C) clayey soil
 - (D) humus
95. Forests keep the soil fertile mainly because:
- (A) animals dig up minerals
 - (B) rocks melt into food
 - (C) trees eat the soil
 - (D) dead leaves and animals decay into humus that enriches the soil

96. $A : B = 5 : 2$. If $B = 6$, then A is:

- (A) 12
- (B) 30
- (C) 15
- (D) 2.4

97. To make bricks that hold their shape, the best soil is:

- (A) sandy soil
- (B) loamy soil
- (C) gravel
- (D) clayey soil

98. Animals that eat BOTH plants and animals are called:

- (A) producers
- (B) carnivores
- (C) herbivores
- (D) omnivores

99. Find 10% of 10% of 1000.

- (A) 10
- (B) 100
- (C) 200
- (D) 1

100. Earthworms and microbes in the soil are helpful because they:

- (A) eat all the crops
- (B) turn soil into rock
- (C) cause soil erosion
- (D) break down dead matter and keep the soil airy and fertile